

Plant diversity effect on water quality in wetlands: a meta-analysis based on experimental systems

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Abstract. The ecological literature reports little empirical evidence from biodiversity–ecosystem functioning (BEF) experiments in wetland systems, even though wetlands are widely known for their water filtering capacity. Experiments comparing the effect of plant monocultures and mixtures on water quality to improve pollutant removal efficiency in treatment wetlands share the characteristics of classical BEF experiments, and so could provide insights for wetland management. To add to our understanding of BEF relationships in wetlands, we evaluated plant diversity effects on water purification through a meta-analysis of freshwater experimental wetlands comparing monocultures to mixtures. We found 28 studies that matched our criteria for BEF analysis, for a total of 561 diversity effects on pollutant removal. Overall, the meta-analysis shows no significant effect of plant richness on removal of total suspended solids, but a positive effect on chemical oxygen demand and total nitrogen removal, and a marginal effect on phosphorus removal. Thus, the results of this meta-analysis are consistent with reports of an overall positive biodiversity effect on ecosystem properties. An analysis of moderator variables shows that the experimental context (size of the experimental units, nutrient load, duration of the experiment) does not explain much of the residual variance. For pollutants that benefit from a positive plant richness effects on removal, mixtures do not perform better than the best monoculture. We found no evidence that plant richness effects are due to functional complementarity among species rather than to the presence of particularly efficient species. Complementarity effects may be less prevalent in highly productive, nutrient-rich wetlands, compared to nutrient-limited environments such as natural grasslands. Although findings must be confirmed by long-term field experiments under natural conditions, result from experimental wetland systems may contribute to a better understanding of biodiversity effect on ecosystem functions in wetlands, in addition to guide practices in natural wetland restoration and the use of constructed wetlands for water treatment.

Key words: biodiversity; determinants of plant community diversity and structure; ecosystem functioning; ecosystem services; plant richness; pollutant removal; treatment wetland.

INTRODUCTION

In the context of climatic and land cover changes, the effect of biodiversity loss on ecosystem functioning has become a central issue in ecology. Biodiversity and ecosystem functioning (BEF) experiments have investigated the response of ecosystem processes to manipulated species richness gradients (Schmid et al. 2002, Bailey and Reiss 2014). Overall conclusions of BEF experiments suggest that there is a positive relationship between biodiversity and ecosystem processes such as biomass production, nutrient recycling, and decomposition (Balvanera et al. 2006, Cardinale et al. 2011, 2012).

The evidence is particularly strong for grassland ecosystems, in which the majority of the BEF experiments were carried out (Cardinale et al. 2011). It has been argued that the loss of biodiversity and its effect on ecosystem functions may, in turn, negatively affect the goods and services that ecosystems provide to humans (Quijas et al. 2010). Contention points concern the relevance of BEF relationships for naturally assembled communities (Wardle 2016), as well as the mechanisms by which biodiversity supports ecosystem functions and services (Cardinale et al. 2012, Winfree et al. 2015).

Wetlands are among the most important ecosystems on Earth for the rich biodiversity they support and the beneficial services they provide (Zedler and Kercher 2005, Mitsch and Gosselink 2007). Wetlands are a major source of food products, fuel and building material, and represent an increasing value for ecotourism. They

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also perform important functions such as carbon sequestration and flood abatement. But one of the most fundamental ecosystem services provided by wetlands is their role in improving water quality. The water purification capacity of wetlands is due to several characteristics, including the high diversity of decomposers and decomposition processes in sediments; the variety of anaerobic and aerobic processes occurring in close proximity that promote chemical reactions; and the high plant productivity, which leads to high rates of mineral uptake by vegetation (Mitsch and Gosselink 2007, Reddy and DeLaune 2008). Wetlands show such a capacity to remove organic and inorganic material flowing across them that they have been described as “the kidney of the world” (Mitsch and Gosselink 2007). Natural wetlands have been used to treat wastewater discharge for more than a century. More recently, treatment wetlands have been developed and constructed specifically for this purpose and are now widely used around the world (Kadlec and Wallace 2009, Dotro et al. 2017).

It has been hypothesized that the high plant diversity of wetlands contributes to their water purifying capacity (Engelhardt and Ritchie 2001, 2002). Species differences in root morphology and seasonal activity may lead to more complete spatial and temporal compensation (Guntenspergen et al. 1989, Coleman et al. 2001, Rheault et al. 2015, Rodriguez and Brisson 2016). Complementarity in resource uses between species may result in increased nutrient uptake and water treatment efficiency (Fraser et al. 2004). Plant species may also differ in the type of environment they provide to microbial communities, and may thus, in combination, lead to greater microbial activity (Zhang et al. 2010, 2011, Bissegger et al. 2014). Niche partitioning among species of algae has been shown to improve water quality through increase in the uptake and storage of nitrate (Cardinale 2011).

The influence of biodiversity on ecosystem services is most easily demonstrated on provisioning and regulating services that can be directly linked to the ecosystem functions measured in BEF experiments (Cardinale et al. 2012). There are strong ties between water quality (a regulating ecosystem service) and ecosystem functions such as nutrient cycling and organic matter decomposition. Thus, wetlands represent an ideal ecosystem to test the effect of species richness on ecosystem service by examining their effect on water treatment and understanding the underlying functions in play. Yet, the ecological literature reports little empirical evidence from experiments conducted in wetlands. For example, in their meta-analysis, Cardinale et al. (2011) suggested, based on data from 59 experiments, that water quality improved as species richness of plants or algae increased (also reported in Balvanera et al. 2014). However, of these experiments, 56 were conducted in temperate grasslands (Fig. 2E in Cardinale et al. 2011). Results found in grasslands may not necessarily apply to forb- or sedge-dominated ecosystems. Managed grasslands are often limited in water supply and soil nutrients (vs.

no water limitation and often high soil nutrients in managed wetlands) and host species with low to moderate biomass, productivity, and specific leaf area (which range from medium to high in wetlands; Shipley and Lechowicz 2000, Craine et al. 2001, Asner et al. 2003). Also, most previous biodiversity experiments measured improvement in water quality through nitrogen uptake, whereas this is only one of the processes involving nitrogen removal from water. The role of species richness in the pathways that lead to long-term storage or permanent removal of nitrogen and other pollutants still requires investigation (Balvanera et al. 2014).

Treatment wetlands technology has evolved considerably in recent decades, with much effort directed towards understanding the factors that influence performance, such as hydraulics and internal biochemical cycling (Dotro et al. 2017). Species identity and diversity in treatment wetlands has also been of interest to researchers specializing in this field, and several experiments comparing water treatment in plant monocultures and mixtures have been conducted (Debing et al. 2010, Rodriguez and Brisson 2016, Türker et al. 2016). In these experiments, model wetlands of various sizes (from small pot-size microcosms to full-size treatment wetlands) and designs (from free water wetlands with floating plants to subsurface flow wetlands with rooted wetland plants) are fed with wastewater of known pollutant concentration. Pollutant removal efficiency is then evaluated by comparing inflow-outflow concentrations. The pollutants considered may include suspended solids, organic matter (most often estimated by chemical oxygen demand and biological oxygen demand), nitrogen (total nitrogen, ammonium, nitrate-nitrite), phosphorus (total phosphorus and phosphate), and, less often, trace elements or emergent pollutants. These experiments are comparable to classical BEF experiments, although they have generally not been conducted in the context of global biodiversity losses, but rather for a practical reason: guiding species selection to improve pollutant removal efficiency in treatment wetlands. Their design (wetland soil and plants, water inlet and outlet, with some variation) is similar from the ones that are used by ecologists to test the relationship between wetland plant richness and water quality in controlled conditions (Engelhardt and Ritchie 2002). Results of these experiments vary, with some of the studies reporting a positive effect of plant richness (Coleman et al. 2001, Wang et al. 2014), while others found no measurable effect (Fraser et al. 2004, Liang et al. 2011), making generalization difficult.

The objective of the present research is to evaluate the effect of plant species richness and composition on water purification in freshwater wetlands through a meta-analysis of experiments that compared monocultures to mixtures. We focus our evaluation on floating and emergent plants because of their predominant direct or indirect effects on wetland processes. More specifically, we will address the following questions: Is there a diversity effect of plants on pollutant removal, and how does it

depend on the type of pollutant and on the experimental design? As this study will demonstrate, we found a positive plant richness effect on some pollutants. Consequently, we asked the following questions: Do wetland mixtures perform better than the best monoculture? Is this diversity effect mostly explained by a complementarity effect or more simply by a plant selection effect? In addition to contributing to fundamental understanding of natural processes, answers to these questions are important for practitioners and decision makers involved in wetland restoration and treatment wetlands.

MATERIALS AND METHODS

Studies selected

In November 2016, we searched the Web of Science database without restriction on publication year and using the following combination of keywords: wetland AND (pollutant removal OR nutrient* removal) AND (plant* species OR macrophyte*) AND (monoculture* OR mixed culture OR polycultures). We also asked colleagues in conferences on constructed wetlands for papers on the subject that would not have been found through our search, as well as references to data in MSc theses or PhD dissertations.

The papers selected for the analysis had to compare plants growing in monocultures and mixtures in experimental wetlands (from floating plants in free water surface wetlands to rooted emergent wetland plants in subsurface flow wetlands). The studies selected had to conform to the following criteria: (1) the identity of the plant species, both in monocultures and mixtures, was clearly established for each experimental unit; (2) the experimental units were closed systems fed with water of known pollutant concentration (inflow), and where pollutant removal could be assessed by monitoring the pollutant concentration of the outflow, or in situ measurements of water quality in the experimental unit after a period of time; (3) removal efficiency (in percent) and standard deviation of at least one of the following common measures of water quality were reported or could be calculated from the data presented in the paper: total phosphorus (TP), total nitrogen (TN), total suspended solids (TSS) or chemical oxygen demand (COD) as a measure of organic matter. Other pollutants were not evaluated because data were too scarce to allow meaningful comparisons.

We did not consider nutrient levels in plant tissues as a measure of water treatment. Plant uptake is only one mechanism responsible for water purification, and the amount of nutrients that can be removed by harvesting is often considered insignificant compared to other processes (Brix 1997).

Data analyses: water treatment and productivity

Data selection.—We built a database of diversity effects in which each row represented a comparison between

the mean removal efficiency (%) of a monoculture of a given species vs. a mixture of species under the same experimental conditions. Because the experiments varied widely in terms of wastewater pollutant load, feeding schedule and mode, as well as design, we used percentage removal as a common measure of treatment efficiency. Percent removal was calculated as $PR = 100 \times (1 - [X_{\text{outflow}}]/[X_{\text{inflow}}])$, where $[X]$ stands for the pollutant concentration in the inflow or outflow. Each study provided data for at least one row, but in general, several rows were entered per study. For example, if a study compared two species as monocultures with the mixture of both species, then two rows were extracted for each pollutant, comparing each monoculture to the mixture of both species. If, in the same study, the diversity effect was tested under different experimental conditions, for example, different nutrient types or loads, each condition was extracted as a separate row.

Diversity effect size and precision.—For each row of the dataset, we calculated a ratio between the outcome of the mixture and the monoculture treatments, and used the natural logarithm of that ratio (hereafter $\log(\text{response ratio})$) as a standardized measure of the effect size. Since each test of a diversity effect was replicated, we calculated the variance of these $\log(\text{response ratios})$ as:

$$V\ln R = S_{\text{pooled}}^2 \times (1/[n1[PR1]^2] + 1/[n2[PR2]^2])$$

where S_{pooled} is the pooled standard deviation among replicates, $n1$ and $n2$ are, respectively, the sample size of the diversity and the monoculture treatment, and $PR1$ and $PR2$ are the percent removal of these treatments (Borenstein et al. 2009). The $\log(\text{response ratios})$ were back-transformed to response ratios for presentation purposes.

Meta-analytic models.—For each diversity effect, we calculated a weighted effect size, using $V\ln R^{-1}$ as weights in a three-level random effects model. Random intercept terms were included for the experiment, as well as the study, within which a variance term was added to account for the identity of the monoculture species.

Between-experiment heterogeneity was quantified using I_R^2 , a multivariate extension of the I^2 metric (Jackson et al. 2012). I_R^2 is on a 0–100% scale and can be interpreted as the proportion of the effect size variance explained by the between-experiment heterogeneity.

For pollutants for which a significant diversity effect was found, this process was repeated on a subset of the data, where only the best monocultures were retained in order to assess if mixtures were more efficient than the best monoculture of the species they contain (an analogue of “transgressive overyielding”; Schmid et al. 2008).

Analysis of moderators.—We conducted an analysis of moderator variables to explore the effects of the experimental context on the log response ratios. For each

experiment, we extracted seven moderator variables that could potentially have an effect on pollutant removal: (1) the scale of the experimental unit (microcosms [$<0.2 \text{ m}^2$], mesocosms [$0.2\text{--}1 \text{ m}^2$], pilot [$1\text{--}10 \text{ m}^2$], and full scale [$>10 \text{ m}^2$]); (2) the nutrient load (in mg/L); (3) the type of wetland flow (horizontal subsurface, vertical subsurface, or free water surface flow); (4) the duration of the experiment (in months); (5) the maturity of the plants at the onset of the experiment (number of months after planting); (6) the type of plants (emergent, free-floating, or submerged); and (7) the maximum species richness tested in mixtures.

We applied a forward-selection procedure to each model, in which, at every step, all moderator variables were independently added to the current best model (beginning with the “empty” meta-analytic models described above). If the moderator with the highest likelihood-ratio value was deemed significant (likelihood-ratio test with $P < 0.05$), we labeled it as an important moderator and added it to the current best model. We repeated these steps until all important moderators had been added to the model. After each step, we calculated variance inflation factors and the variable selection process was stopped if the newly added moderator moved inflation factors above 10 (Neter et al. 1989).

The presence of a publication bias in the reviewed literature was assessed with the regression test for funnel plots (Sterne and Egger 2005) and the trim-and-fill method using the R_0 estimator (Duval and Tweedie 2000). All the above data analyses were conducted with the meta for package (version 1.9-9; Viechtbauer 2010) in R statistical software (version 3.3.2; R Core Team 2016).

RESULTS

We found 28 studies that matched our criteria (Table 1). Most papers presenting diversity effects that were not selected either lacked identification of each individual species in their mixtures, studied other pollutants than the ones targeted for the meta-analysis, or were not conducted in closed experimental units. The number of species in the mixtures ranged from 2 to 6 (Fig. 1). The most common species represented in the experiments were *Typha latifolia* (seven studies), *Phragmites australis* (six studies), *Eichornia crassipes* (six studies), and *Phalaris arundinacea* (five studies). In one study (Tomimatsu et al. 2014), six genotypes of *Phragmites australis* were studied rather than six different species per se. The sizes of the experimental units varied from small microcosms of 0.02 m^2 to large full-size artificial wetlands of $1,000 \text{ m}^2$. However, the surface area of a majority of the experimental units was less than 4 m^2 (Fig. 1). There was often an inverse relationship between the size of the experimental units and the number of replicates, most likely due to the high cost of running experiments in large units.

Pollutant removal varied from 19% to 99% (Table 2, Fig. 2) and most often reached above 50%, independently

of the pollutant or the study. TN removal was the most common pollutant removal monitored (22 studies). The total number of comparisons between monocultures and mixtures ranges from 66 (TSS removal) to 215 (TP removal) (Table 2). In total, our database contains 561 diversity effects on pollutant removal efficiency.

The comparison of pollutant removal from mixtures and monocultures indicates that there is a significant but low diversity effect on TN and COD removal, a marginal effect on TP removal and no effect on TSS removal (Fig. 3). The average response ratio for pollutant removal is low (<1.1). However, low values are inevitable for mathematical reasons because we expressed pollutant removal as percentages. Removal in monocultures is typically higher, and sometimes much higher, than 50%, which sets a narrow upper limit to the proportional increase that can be attributed to plant richness. In all cases, high amounts of between-experiment heterogeneity were observed (all I_R^2 values are $>98\%$), suggesting that diversity effects might be highly context specific (Table 3). Although potentially important moderators were identified for some pollutants based on

TABLE 1. Studies selected for the meta-analysis of the plant diversity effect on water treatment of different pollutants: total phosphorus (TP), total nitrogen (TN), chemical oxygen demand (COD) and total suspended solids (TSS).

Experimental studies	Pollutant removal efficiency			
	TP	TN	COD	TSS
Arroyave (2010)			X	
Bachand and Horne (1999)		X		
Coleman et al. (2001)		X	X	X
Dai et al. (2014)	X	X	X	
Debing et al. (2009)	X	X	X	
Debing et al. (2010)	X	X	X	
Ellerton et al. (2012)	X	X		X
Engelhardt and Ritchie (2002)	X			
Fraser et al. (2004)	X	X		
Han et al. (2016)		X		
Karathanasis et al. (2003)			X	X
Karpiscak et al. (1996)		X	X	
Kumari and Tripathi (2014)	X	X	X	
Liang et al. (2011)	X	X	X	
Lindnermer (2015)	X	X		
Menon and Holland (2014)	X			
Picard et al. (2005)	X	X		
Prajapati et al. (2013)			X	X
Rodriguez (2014)	X	X	X	X
Rodriguez and Brisson (2016)	X	X	X	X
Sooknah and Wilkie (2004)	X	X	X	X
Tomimatsu et al. (2014)		X		
Tripathi and Upadhyay (2003)	X	X		
Türker et al. (2016)		X		
Wang et al. (2014)		X		
Zhang et al. (2007)	X	X		
Zhao et al. (2014)	X	X		
Zurita et al. (2009)	X	X	X	X

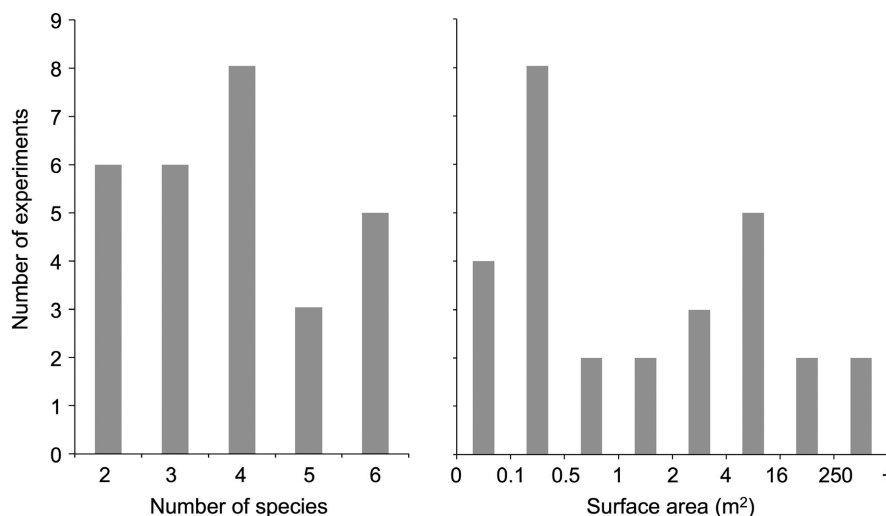


FIG. 1. Distribution of the maximum number of species and of the sizes (surface area) of experimental units in the 28 selected studies.

likelihood-ratio tests, none of them explained much of the residual variance (Table 3).

While there are indications of a biodiversity effect on TN and COD removal, this effect does not lead into mixtures being more efficient than their best monoculture (Fig. 4).

A publication bias was detected for two of the four measures (TP and TN; see Appendix S1: Table S1). The trim-and-fill method revealed that, for each measure, a few studies were missing to the left of the funnel (Appendix S1: Fig. S1). Such an asymmetry suggests a slight bias towards detecting a positive diversity effect. However, this publication bias is not particularly worrisome because most of the “missing studies” identified by the trim-and-fill method have a high standard error (Appendix S1: Fig. S1), and thus, would be strongly down-weighted in the meta-analyses.

DISCUSSION

Overall, our meta-analysis suggests that plant richness does have a low but positive effect on COD and TN removal and marginally on TP removal. The fact that diverse systems were no more efficient than the monoculture with the best species they include suggests that

this diversity effect is more likely due to a selection effect than to a complementarity between species.

Pollutant removal in wetlands involves complex physical, chemical and biological processes (Reddy and DeLaune 2008). Yet, plant richness can only foster removal of a specific pollutant if plant presence, irrespective of richness, indeed plays a role in water purification for that pollutant. The absence of a diversity effect on TSS and the positive effects on COD and TN removal found in our meta-analysis are consistent with the recognized role of plants in wetlands. TSS removal is mainly the result of physical processes of sedimentation and filtration through the substrate. The plants may play a role by preventing particles resuspension (Kadlec and Wallace 2009). However, these processes may not occur in smaller experimental units, or even have negative impacts over time (Knowles et al. 2011). COD and TN removal occur primarily through microbiological degradation, and plants are important in stimulating microbiological processes by providing attachment surfaces for microbial growth as well as releasing oxygen and exudates in the rhizosphere (Brix 1997, Faulwetter et al. 2009). The majority of experiments that compared planted vs. unplanted artificial wetlands showed that the formers were more efficient in pollutant removal, and that the most common benefit pertained to nitrogen removal (Tanner 2001, Brisson and Chazarenc 2009). In wetlands, where the main mechanisms for long-term TP removal is through soil adsorption, plant diversity is not expected to significantly influence removal. The vegetation may serve as a short-term sink through plant uptake, but this phosphorus is released at death through decomposition (Richardson 1985). Thus, a positive effect of plants may be measured in short-term experiments conducted during the growing season (Tanner 2001), as was the case for many experiments in our

TABLE 2. Range of results for pollutant removal, number of studies, and diversity effects reported.

Parameter	Units	Range	No. studies	No. measures of diversity effect
TP removal	%	22–99	18	215
TN removal	%	19–99	23	199
COD removal	%	37–98	14	81
TSS removal	%	19–98	8	66

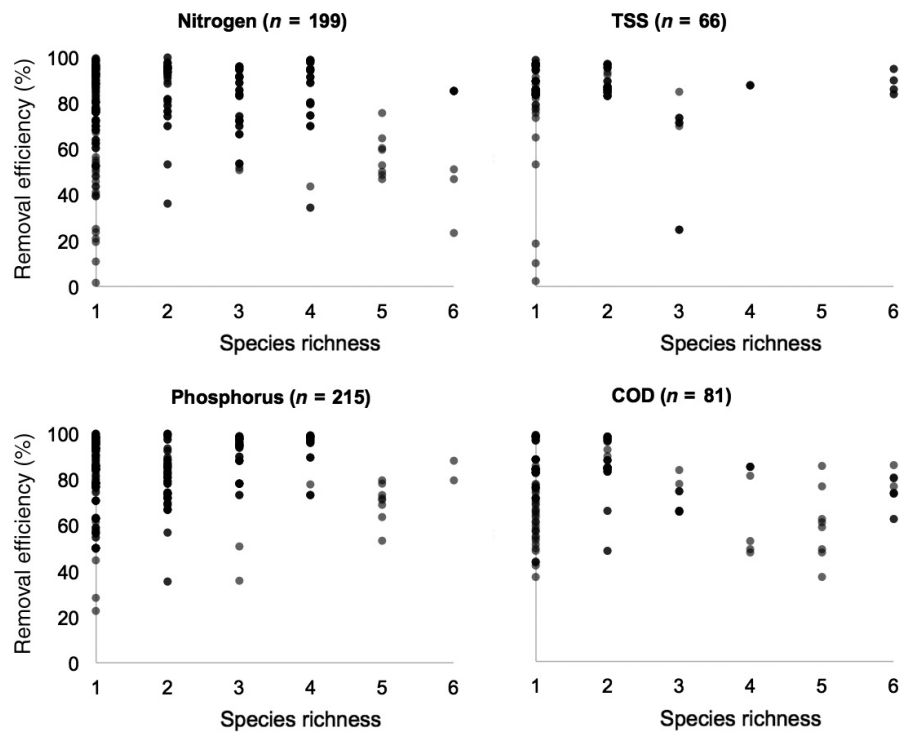


FIG. 2. Scatter plot of pollutant removal efficiency in relation to plant species richness for the 561 diversity effects for total phosphorus, total nitrogen, chemical oxygen demand (COD), and total suspended solids (TSS). Overlapping points are solid black.

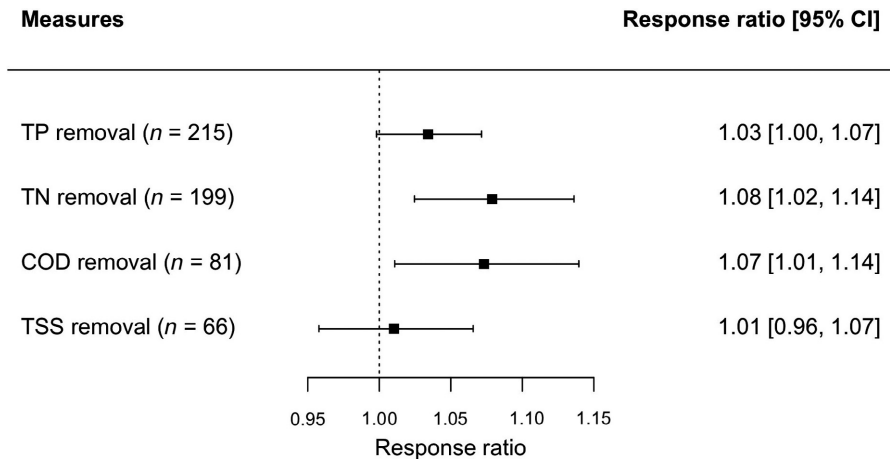


FIG. 3. Response ratios and 95% confidence intervals of the removal efficiency of mixtures relative to monocultures for total phosphorus (TP), total nitrogen (TN), chemical oxygen demand (COD), and total suspended solids (TSS).

meta-analysis, which may explain the marginal diversity effect we found. However, over a longer period, much of this phosphorus would be released later through decomposition processes, cancelling any possible benefits of plant diversity.

Several experiments conducted in artificial treatment wetlands, but excluded from our meta-analysis because they failed to meet our criteria, reported a plant diversity effect on ecological processes. Zhang et al. (2012a) and

Ge et al. (2015) both found that plant diversity positively affected nitrogen removal in experimental wetlands. Nitrification and denitrification rates were found to increase under higher plant richness (Chang et al. 2014, Zhang et al. 2016), although other studies have reported the opposite trend (Korol et al. 2016). Methane emission was also found to increase with plant richness (Zhang et al. 2012b, Wang et al. 2013, Niu et al. 2015), while N₂O emissions either increased or decreased

TABLE 3. Residual deviance explained by the moderator variables retained in the model.

Parameter	Tau ²	Rho	Important moderators	Residual deviance explained by moderators
TP removal	0.005	0.508	flow type, scale	2.6%
TN removal	0.005	0.691	species richness, plant type, maturity	1.1%
COD removal	0.012	0.977	—	—
TSS removal	0.001	0.345	—	—

Note: —, no residual deviance.

depending on experimental conditions (Chang et al. 2014, Han et al. 2016, 2017). The positive effect of plant diversity on biomass productivity, a measure of carbon sequestration, was also reported in several experiments not reviewed in our meta-analysis (Chang et al. 2014, Zhang et al. 2016, Zhao et al. 2016), although others failed to find an effect (Korol et al. 2016, Luo et al. 2016, Means et al. 2016).

The results of this meta-analysis are consistent with reports of an overall positive biodiversity effect on ecosystem properties (Balvanera et al. 2006). However, we found no evidence that these effects are related to “niche complementarity.” Theoretically, an ecosystem with greater plant richness would be expected to display a wider range of growth strategies and functional traits, thus increasing resource use efficiency and productivity (Díaz and Cabido 2001, Loreau et al. 2001). Alternatively, the positive effect of plant richness on treatment efficiency in our meta-analysis could simply be caused by a “sampling effect,” which is a greater probability of including individual species with better performance (Loreau and Hector 2001). The sampling effect reflects the increasing probability that the best performing species would be present in mixtures and have a

disproportionate influence on ecosystem processes (Tilman 1999, Loreau and Hector 2001).

Complementarity effects may be more important in nutrient-limited environments, such as natural grasslands, where the competition for resources is symmetric (Cardinale et al. 2011, DeMalach et al. 2016). In contrast, natural wetlands are among the most productive ecosystems, in part because they receive sediments and nutrients from the drained watershed (Reddy and DeLaune 2008). Similarly, experimental wetlands in our meta-analysis were fed with artificial polluted water or a nutrient solution, so that they were not nutrient limited. The plant species typically used in experiments are those already known to perform well in treatment wetlands. Consequently, a complementarity effect on water treatment efficiency is less likely in experimental than in natural wetlands. In an experimental context, complementarity and selection effects can be distinguished when the measured diversity effect can be partitioned among species, as is the case for biomass or nutrient levels in plant tissues. However, partitioning the contribution of each species to pollutant removal in a mixture is not possible with the available data. Furthermore, the experiments evaluated here were conducted with no more than six species and rarely with all possible combinations, which prevented the detection of species identity effects.

Overall, we found no evidence to support the hypothesis that plant mixtures are more efficient in pollutant removal than their most efficient monoculture. “Transgressive overyielding” occurs when a polyculture yields more biomass than the monoculture of its best yielding species (Trenbath 1974). Transgressive overyielding and its analogues on other ecological functions are often considered as an unequivocal evidence for complementarity effect (Loreau 1998, Cardinale et al. 2011). Conversely, when species identity is the main driver of the diversity effect, the ecological function in a mixture can never be higher than that of its most efficient species

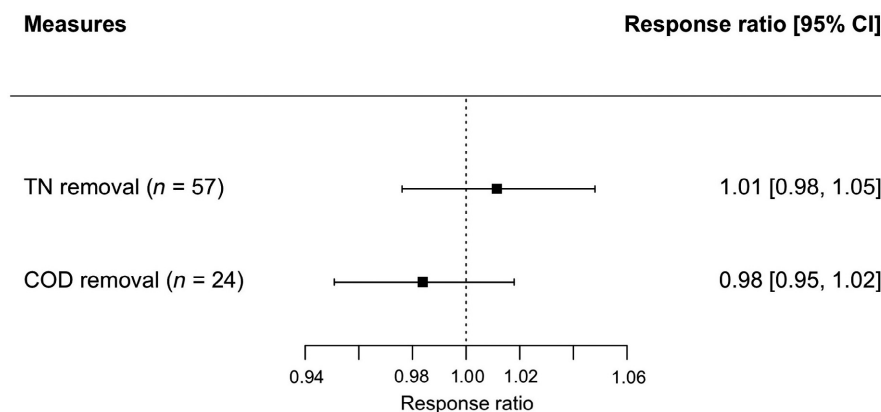


FIG. 4. Response ratios and 95% confidence intervals of the removal efficiency of mixtures relative to the best monocultures for total nitrogen (TN) and chemical oxygen demand (COD).

grown in a monoculture. A literature review conducted by Cardinale et al. (2007) reports that transgressive overyielding is generally stronger when experiments run longer. Most studies in our meta-analysis were short-term experiments, so that the analog of transgressive overyielding for water quality may have been underestimated, although study duration was not an important moderator of the diversity effect. The absence of an analog of transgressive overyielding in TN or COD removal further argues in favor of a selection effect. Because competition among plant species is typically strong in wetlands, often resulting in local dominance of one or a few species, it has been suggested that sampling effects will tend to dominate the relationship between aquatic plant species richness and ecosystem functioning (Engelhardt and Ritchie 2002). In wetland studies where a positive relation between water treatment efficiency and plant richness was reported, the mechanisms at work is occasionally presumed to be a complementarity between species, with little consideration for the alternatives (Karathanasis et al. 2003, Zhang et al. 2012a).

There is a large body of evidence linking biodiversity and functions, especially for productivity. This positive relation has been particularly well established for grasslands (Cardinale et al. 2011), but also, more recently, for forest ecosystems (Paquette and Messier 2011, Liang et al. 2016). There is growing evidence that this positive effect of diversity on functions translates into ecosystem services such as forage production, timber production, and climate regulations (Balvanera et al. 2014). As for water quality services, a review covering 59 experiments (56 in grasslands, 2 in wetlands, 1 in a lake) found an overall positive diversity effect in nutrient reduction in soil or water (Cardinale et al. 2011, Balvanera et al. 2014). However, similar to our own results on pollutant reduction in wetlands, the reduction in nutrients was generally higher in the most efficient monoculture (Cardinale et al. 2011).

Experiments conducted with artificial, simplified, model ecosystems under controlled conditions have greatly contributed and continue to contribute to ecological knowledge. Yet, one may question whether our results can be extrapolated to natural wetlands, or apply only to experimental systems. It is worth emphasizing that the same question can be raised about experimental grassland and forest plantation experiments that manipulated species composition. In a recent study of two of the longest-running biodiversity experiments (Jena Experiment, Germany; BioDIV, USA), the authors re-analyzed a subset of grassland communities that are comparable to real-world communities. For most biodiversity–ecosystem–functioning relationships, biodiversity effects were still detectable and similar in magnitude (Jochum et al. 2019). However, it must be pointed out that most diversity experiments do not take spatial heterogeneity into account. The biodiversity experiments from our meta-analysis, like most biodiversity experiments, were conducted under completely homogenous

conditions to better isolate the effect of plant diversity alone from possible confounding factors such as the direct or indirect effects of spatial heterogeneity on pollutant removal. In natural wetlands, plant diversity is highly related to spatial heterogeneity in water level, nutrient availability or salinity gradient (Pollock et al. 1998, Raulings et al. 2010, Janousek and Folger 2014). The higher plant diversity may, in turn, increase ecosystem function and services such as water quality. This pattern was shown for freshwater algae in a stream mesocosm experiment where systems with greater heterogeneity in flow and disturbance had higher algae diversity and higher nitrate uptake through niche differentiation (Cardinale 2011). As mentioned by Balvanera et al. (2006), extrapolation of small-scale experiments to the larger landscape scale is limited by the greater environmental heterogeneity and its effects on ecosystem functioning. Future studies should investigate the relative contribution of plant diversity and spatial, or temporal, heterogeneity in wetland ecosystem services and functions (Sutton-Grier et al. 2011).

Several experiments used in our meta-analysis were done in mesocosms. This approach has the same advantages (replication, mechanistic insight, control over the factor of interest: here, plant richness) and limitations (edge-effects, artificiality) than those of the numerous experiments that have so greatly benefited ecological theory (Passioura 2006, Freckleton et al. 2009). The tight confinement inherent to mesocosms may affect processes even more when it involves interaction between several plant species. Experimental treatment wetlands are labor intensive, so that there must often be a compromise between the number of units and their sizes. Also, compared to most BEF experiments, experimental wetlands use a smaller species pool. Yet, because a positive but saturating curve is the shape that predominantly describes the diversity–function relationship derived from biodiversity experiments (Hector and Bagchi 2007, Cardinale et al. 2011), a low number of species should not impede the ability to detect positive relationship, but only how this relationship may decline at greater diversity levels. Moreover, analysis of moderator variables in our meta-analytic model suggests that experimental conditions (number of units, number of species, and duration) do not explain the residual variance. Yet, our findings must be confirmed by observational evidence or long-term field experiments conducted under natural conditions. Meta-analysis on wetland restoration showed that ecosystem structure and function recovery is very slow (Moreno-Mateos et al. 2012, 2015) and while restoration may improve ecosystem services (including water quality) and wetland biodiversity (Meli et al. 2014), no causal connection was made between the two.

The number of plant diversity experiments in experimental wetlands has been growing over the last decade, and it must be hoped that a better integration with ecological theory will help broaden our understanding of

ecosystem processes. While this meta-analysis does not provide evidence of functional complementarity, some experiments do suggest that it may occur under some circumstances (Picard et al. 2005, Rodriguez 2014). The impact of plant diversity on microbial activity and diversity in a wetland rhizosphere may affect nutrient recycling (Sutton-Grier et al. 2013, Wang et al. 2013). More experiments are needed to better understand the circumstances that favor complementarity vs. selection effects, and their relative impact on nutrient removal. One aspect that is particularly promising is the evaluation of how plant diversity affects the removal of many pollutants, rather than each one considered individually. For example, in one experiment with two species, we showed that combining both species would not improve treatment efficiency for any pollutant taken individually; i.e., the best monoculture was always as efficient as or more efficient at removal than the combination of two plant species. However, combining both plant species represented the best trade-off between overall high pollutant removal and low nitrate levels in the effluent (Rodriguez and Brisson 2016). These results raise the more general, yet understudied, question of how plant diversity may simultaneously affect several ecological functions, the so-called “ecosystem multi-functionality” (Hector and Bagchi 2007, Gamfeldt et al. 2008, Manning et al. 2018). In treatment wetlands, the processes for removing pollutants differ from one to another. Consequently, treatment wetland experiments offer an ideal situation for testing hypotheses on the multi-functionality of bio-diversified ecosystems.

The results of our meta-analysis not only contribute to ecological theory, they also have practical consequences in treatment wetland design. Our findings suggest that a more diverse restored or treatment wetland containing a highly effective species may be no more efficient in water quality improvement than a monospecific stand of that species. Alternatively, the “selection effect” is such that there is no evidence that using a combination of species is less efficient than using only its most efficient one. Thus, when a range of plant species is available for designing treatment wetlands, but no information exists on their respective efficiency, a combination of plant species may be the best option to insure maximum performance. In addition, greater plant richness in treatment wetlands may foster the provision of auxiliary services like high habitat value for fauna, resilience to environmental stressors or diseases, as well as high esthetical value, without compromising water treatment efficiency.

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LITERATURE CITED

- Arroyave, A. 2010. Efectos del tipo de vegetación y de las variaciones de profundidad en la eficiencia de remoción de patógenos en humedales construidos de flujo subsuperficial. Thesis. Universidad Tecnológica de Pereira, Pereira, Colombia.
- Asner, G. P., J. M. O. Scurlock, and J. A. Hicke. 2003. Global synthesis of leaf area index observations: implications for ecological and remote sensing studies. *Global Ecology and Biogeography* 12:191–205.
- Bachand, P. A., and A. J. Horne. 1999. Denitrification in constructed free-water surface wetlands: II. Effects of vegetation and temperature. *Ecological Engineering* 14:17–32.
- Bailey, R. A., and J. Reiss. 2014. Design and analysis of experiments testing for biodiversity effects in ecology. *Journal of Statistical Planning and Inference* 144:69–80.
- Balvanera, P., A. B. Pfisterer, N. Buchmann, J. S. He, T. Nakashizuka, D. Raffaelli, and B. Schmid. 2006. Quantifying the evidence for biodiversity effects on ecosystem functioning and services. *Ecology Letters* 9:1146–1156.
- Balvanera, P., I. Siddique, L. Dee, A. Paquette, F. Isbell, A. Gonzalez, J. Byrnes, M. I. O'Connor, B. A. Hungate, and J. N. Griffin. 2014. Linking biodiversity and ecosystem services: current uncertainties and the necessary next steps. *BioScience* 64:49–57.
- Bisseger, S., M. Rodriguez, J. Brisson, and K. P. Weber. 2014. Catabolic profiles of microbial communities in relation to plant identity and diversity in free-floating plant treatment wetland mesocosms. *Ecological Engineering* 67:190–197.
- Borenstein, M., L. V. Hedges, J. P. T. Higgins, and H. R. Rothstein. 2009. Introduction to meta-analysis. John Wiley and Sons, Chichester, UK.
- Brisson, J., and F. Chazarenc. 2009. Maximizing pollutant removal in constructed wetlands: Should we pay more attention to macrophyte species selection? *Science of the Total Environment* 407:3923–3930.
- Brix, H. 1997. Do macrophytes play a role in constructed treatment wetlands? *Water Science and Technology* 35:11–17.
- Cardinale, B. J. 2011. Biodiversity improves water quality through niche partitioning. *Nature* 472:86–89.
- Cardinale, B. J., K. L. Matulich, D. U. Hooper, J. E. Byrnes, E. Duffy, L. Gamfeldt, P. Balvanera, M. I. O'Connor, and A. Gonzalez. 2011. The functional role of producer diversity in ecosystems. *American Journal of Botany* 98:572–592.
- Cardinale, B. J., J. P. Wright, M. W. Cadotte, I. T. Carroll, A. Hector, D. S. Srivastava, M. Loreau, and J. J. Weis. 2007. Impacts of plant diversity on biomass production increase through time because of species complementarity. *Proceedings of the National Academy of Sciences USA* 104:18123–18128.
- Cardinale, B. J., et al. 2012. Biodiversity loss and its impact on humanity. *Nature* 486:59–67.
- Chang, J., et al. 2014. Plant species richness enhances nitrous oxide emissions in microcosms of constructed wetlands. *Ecological Engineering* 64:108–115.
- Coleman, J., K. Hench, K. Garbutt, A. Sexstone, G. Bissonnette, and J. Skousen. 2001. Treatment of domestic

- wastewater by three plant species in constructed wetlands. *Water, Air, and Soil Pollution* 128:283–295.
- Craine, J. M., J. Froehle, D. G. Tilman, D. A. Wedin, and F. S. Chapin III. 2001. The relationships among root and leaf traits of 76 grassland species and relative abundance along fertility and disturbance gradients. *Oikos* 93:274–285.
- Dai, Y., H. Tang, J. Chang, Z. Wu, and W. Liang. 2014. What's better, *Ceratophyllum demersum* L. or *Myriophyllum verticillatum* L., individual or combined? *Ecological Engineering* 70:397–401.
- Debing, J., S. Baoqing, Z. Hong, and H. Jianming. 2010. Chemical oxygen demand, nitrogen and phosphorus removal by subsurface wetlands with *Phragmites* vegetation in different models. *Engineering in Life Sciences* 10:177–183.
- Debing, J., Z. Lianbi, Y. Xiaosong, H. Jianming, Z. Mengbin, and W. Yuzhong. 2009. COD, TN and TP removal of *Typha* wetland vegetation of different structures. *Polish Journal of Environmental Studies* 18:183–190.
- DeMalach, N., E. Zaady, J. Weiner, and R. Kadmon. 2016. Size asymmetry of resource competition and the structure of plant communities. *Journal of Ecology* 104:899–910.
- Díaz, S., and M. Cabido. 2001. Vive la difference: plant functional diversity matters to ecosystem processes. *Trends in Ecology and Evolution* 16:646–655.
- Dotro, G., G. Langergraber, P. Molle, J. Nivala, J. Puigagut, O. R. Stein, and M. von Sperling. 2017. Treatment wetlands. Biological wastewater treatment series. IWA Publishing, London, UK.
- Duval, S. J., and R. L. Tweedie. 2000. Trim and fill: A simple funnel-plot-based method of testing and adjusting for publication bias in meta-analysis. *Biometrics* 56:455–463.
- Ellerton, J. P., B. E. Hatt, and T. D. Fletcher. 2012. Mixed plantings of *Carex appressa* and *Lomandra longifolia* improve pollutant removal over a monoculture of *L. longifolia* in stormwater biofilters Water sensitive urban design; Building the water sensitive community. Pages 164–170 in Engineers Australia, editor. 7th International Conference on Water Sensitive Urban Design [online]. Melbourne Cricket Ground, Barton, Australian Capital Territory, Australia.
- Engelhardt, K. A., and M. E. Ritchie. 2001. Effects of macrophyte species richness on wetland ecosystem functioning and services. *Nature* 411:687–689.
- Engelhardt, K. A., and M. E. Ritchie. 2002. The effect of aquatic plant species richness on wetland ecosystem processes. *Ecology* 83:2911–2924.
- Faulwetter, J. L., V. Gagnon, C. Sundberg, F. Chazarenc, M. D. Burr, J. Brissou, A. K. Camper, and O. R. Stein. 2009. Microbial processes influencing performance of treatment wetlands: a review. *Ecological Engineering* 35:987–1004.
- Fraser, L. H., S. M. Carty, and D. Steer. 2004. A test of four plant species to reduce total nitrogen and total phosphorus from soil leachate in subsurface wetland microcosms. *Biore-source Technology* 94:185–192.
- Freckleton, R. P., A. R. Watkinson, and M. Rees. 2009. Measuring the importance of competition in plant communities. *Journal of Ecology* 97:379–384.
- Gamfeldt, L., H. Hillebrand, and P. R. Jonsson. 2008. Multiple functions increase the importance of biodiversity for overall ecosystem functioning. *Ecology* 89:1223–1231.
- Ge, Y., et al. 2015. Positive effects of plant diversity on nitrogen removal in microcosms of constructed wetlands with high ammonium loading. *Ecological Engineering* 82:614–623.
- Guntenspergen, G. R., F. Stearns, and J. A. Kadlec. 1989. Wetland vegetation. Pages 73–88 in D. A. Hammer, editor. Constructed wetlands for wastewater treatment: municipal, industrial and agricultural. Lewis Publisher, Chelsea, Michigan, USA and CRC Press, Boca Raton, Florida, USA.
- Han, W., J. Chang, X. Fan, Y. Du, S. X. Chang, C. Zhang, and Y. Ge. 2016. Plant species diversity impacts nitrogen removal and nitrous oxide emissions as much as carbon addition in constructed wetland microcosms. *Ecological Engineering* 93:144–151.
- Han, W., M. Shi, J. Chang, Y. Ren, R. Xu, C. Zhang, and Y. Ge. 2017. Plant species diversity reduces N₂O but not CH₄ emissions from constructed wetlands under high nitrogen levels. *Environmental Science and Pollution Research*. <https://doi.org/10.1007/s11356-016-8288-3>
- Hector, A., and R. Bagchi. 2007. Biodiversity and ecosystem multifunctionality. *Nature* 448:188.
- Jackson, D., I. R. White, and R. D. Riley. 2012. Quantifying the impact of between-study heterogeneity in multivariate meta-analyses. *Statistics in Medicine* 31:3805–3820.
- Janousek, C. N., and C. L. Folger. 2014. Variation in tidal wetland plant diversity and composition within and among coastal estuaries: assessing the relative importance of environmental gradients. *Journal of Vegetation Science* 25:534–545.
- Jochum, M., et al. 2019. The results of biodiversity-ecosystem functioning experiments are realistic. *bioRxiv* 725812. <https://doi.org/10.1101/725812>
- Kadlec, R. H., and S. Wallace. 2009. Treatment wetlands. CRC Press Taylor and Francis Group, Boca Raton, Florida, USA.
- Karathanasis, A. D., C. L. Potter, and M. S. Coyne. 2003. Vegetation effects on fecal bacteria, BOD, and suspended solid removal in constructed wetlands treating domestic wastewater. *Ecological Engineering* 20:157–169.
- Karpiscak, M. M., C. P. Gerba, P. M. Watt, K. E. Foster, and J. A. Falabi. 1996. Multi-species plant systems for wastewater quality improvements and habitat enhancement. *Water Science and Technology* 33:231–236.
- Knowles, P., G. Dotro, J. Nivala, and J. García. 2011. Clogging in subsurface-flow treatment wetlands: occurrence and contributing factors. *Ecological Engineering* 37:99–112.
- Korol, A. R., C. Ahn, and G. B. Noe. 2016. Richness, biomass, and nutrient content of a wetland macrophyte community affect soil nitrogen cycling in a diversity-ecosystem functioning experiment. *Ecological Engineering* 95:252–265.
- Kumari, M., and B. D. Tripathi. 2014. Effect of aeration and mixed culture of *Eichhornia crassipes* and *Salvinia natans* on removal of wastewater pollutants. *Ecological Engineering* 62:48–53.
- Liang, M. Q., C. F. Zhang, C. L. Peng, Z. L. Lai, D. F. Chen, and Z. H. Chen. 2011. Plant growth, community structure, and nutrient removal in monoculture and mixed constructed wetlands. *Ecological Engineering* 37:309–316.
- Liang, J., et al. 2016. Positive biodiversity-productivity relationship predominant in global forests. *Science* 354:aaf8957.
- Lindermer, M. 2015. Retention von Stickstoff und Phosphor in experimentellen Mono- und Mischkulturen. Thesis. University of Freiburg, Freiburg, Germany.
- Loreau, M. 1998. Separating sampling and other effects in biodiversity experiments. *Oikos* 82:600–602.
- Loreau, M., and A. Hector. 2001. Partitioning selection and complementarity in biodiversity experiments. *Nature* 412:72–76.
- Loreau, M., et al. 2001. Biodiversity and ecosystem functioning: current knowledge and future challenges. *Science* 294:804–808.
- Luo, B., Y. Ge, W. Han, X. Fan, Y. Ren, Y. Du, M. Shi, and J. Chang. 2016. Decreases in ammonia volatilization in response to greater plant diversity in microcosms of constructed wetlands. *Atmospheric Environment* 142:414–419.
- Manning, P., F. Plas, S. Soliveres, E. Allan, F. T. Maestre, G. Mace, M. J. Whittingham, and M. Fischer. 2018. Redefining

- ecosystem multifunctionality. *Nature Ecology & Evolution* 2:427.
- Means, M. M., C. Ahn, A. R. Korol, and L. D. Williams. 2016. Carbon storage potential by four macrophytes as affected by planting diversity in a created wetland. *Journal of Environmental Management* 165:133–139.
- Meli, P., J. M. R. Benayas, P. Balvanera, and M. M. Ramos. 2014. Restoration enhances wetland biodiversity and ecosystem service supply, but results are context-dependent: a meta-analysis. *PLoS ONE* 9:e93507.
- Menon, R., and M. M. Holland. 2014. Phosphorus release due to decomposition of wetland plants. *Wetlands* 34:1191–1196.
- Mitsch, W. J., and J. G. Gosselink. 2007. *Wetlands*. Fourth edition. John Wiley and Sons, New York, New York, USA.
- Moreno-Mateos, D., P. Meli, M. I. Vara-Rodríguez, and J. Aronson. 2015. Ecosystem response to interventions: lessons from restored and created wetland ecosystems. *Journal of Applied Ecology* 52:1528–1537.
- Moreno-Mateos, D., M. E. Power, F. A. Comín, and R. Yockteng. 2012. Structural and functional loss in restored wetland ecosystems. *PLoS Biology* 10:e1001247.
- Neter, J., W. Wasserman, and M. H. Kutner. 1989. *Applied linear regression models*. Irwin, Homewood, Illinois, USA.
- Niu, C., X. He, Y. Ge, J. Chang, and Z. Lu. 2015. Effect of plant species richness on methane fluxes and associated microbial processes in wetland microcosms. *Ecological Engineering* 84:250–259.
- Paquette, A., and C. Messier. 2011. The effect of biodiversity on tree productivity: from temperate to boreal forests. *Global Ecology and Biogeography* 20:170–180.
- Passioura, J. B. 2006. The perils of pot experiments. *Functional Plant Biology* 33:1075–1079.
- Picard, C. R., L. H. Fraser, and D. Steer. 2005. The interacting effects of temperature and plant community type on nutrient removal in wetland microcosms. *Bioresource Technology* 96:1039–1047.
- Pollock, M. M., R. J. Naiman, and T. A. Hanley. 1998. Plant species richness in riparian wetlands—a test of biodiversity theory. *Ecology* 79:94–105.
- Prapapati, U. B., A. K. Dwivedi, and S. Singh. 2013. Sustainable utilization of aquatic macrophyte for waste water treatment. *Advances in Plant Sciences* 26:443–446.
- Quijas, S., B. Schmid, and P. Balvanera. 2010. Plant diversity enhances provision of ecosystem services: a new synthesis. *Basic and Applied Ecology* 11:582–593.
- R Core Team. 2016. R: a language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria.
- Raulings, E. J., K. A. Y. Morris, M. C. Roache, and P. I. Boon. 2010. The importance of water regimes operating at small spatial scales for the diversity and structure of wetland vegetation. *Freshwater Biology* 55:701–715.
- Reddy, K. R., and R. D. DeLaune. 2008. *Biogeochemistry of wetlands: science and applications*. CRC Press, Boca Raton, Florida, USA.
- Rheault, G., R. Proulx, and L. Bonin. 2015. Plant species richness prolongs the growing season of freely assembled riparian herbaceous communities under dry climatic conditions. *Agriculture, Ecosystems and Environment* 200:71–78.
- Richardson, C. J. 1985. Mechanisms controlling phosphorus retention capacity in freshwater wetlands. *Science* 228:1424–1427.
- Rodriguez, M. 2014. Influence of plant species and richness on pollutant removal in treatment wetlands. Dissertation. Montreal University, Montreal, Quebec, Canada.
- Rodriguez, M., and J. Brisson. 2016. Does the combination of two plant species improve removal efficiency in treatment wetlands? *Ecological Engineering* 91:302–309.
- Schmid, B., A. Hector, M. A. Huston, P. Inchausti, I. Nijs, P. W. Leadley, and D. Tilman. 2002. The design and analysis of biodiversity experiments. Pages 61–75 in M. Loreau, S. Naeem, and P. Inchausti, editors. *Biodiversity and ecosystem functioning: synthesis and perspectives*. Oxford University Press, Oxford, UK.
- Schmid, B., A. Hector, P. Saha, and M. Loreau. 2008. Biodiversity effects and transgressive overyielding. *Journal of Plant Ecology* 1:95–102.
- Shipley, B., and M. J. Lechowicz. 2000. The functional co-ordination of leaf morphology, nitrogen concentration, and gas exchange in 40 wetland species. *Ecoscience* 7:183–194.
- Sooknah, R. D., and A. C. Wilkie. 2004. Nutrient removal by floating aquatic macrophytes cultured in anaerobically digested flushed dairy manure wastewater. *Ecological Engineering* 22:27–42.
- Sterne, J. A. C., and M. Egger. 2005. Regression methods to detect publication and other bias in meta-analysis. Pages 99–110 in H. R. Rothstein, A. J. Sutton, and M. Borenstein, editors. *Publication bias in meta-analysis: prevention, assessment, and adjustments*. Wiley, Chichester, UK.
- Sutton-Grier, A. E., J. P. Wright, B. M. McGill, and C. Richardson. 2011. Environmental conditions influence the plant functional diversity effect on potential denitrification. *PLoS ONE* 6:e16584.
- Sutton-Grier, A. E., J. P. Wright, and C. J. Richardson. 2013. Different plant traits affect two pathways of riparian nitrogen removal in a restored freshwater wetland. *Plant and Soil* 365:41–57.
- Tanner, C. C. 2001. Plants as ecosystem engineers in subsurface-flow treatment wetlands. *Water Science and Technology* 44:9–17.
- Tilman, D. 1999. The ecological consequences of changes in biodiversity: a search for general principles. *Ecology* 80:1455–1474.
- Tomimatsu, H., K. Nakano, N. Yamamoto, and Y. Suyama. 2014. Effects of genotypic diversity of *Phragmites australis* on primary productivity and water quality in an experimental wetland. *Oecologia* 175:163–172.
- Trenbath, B. R. 1974. Biomass productivity in mixtures. *Advances in Agronomy* 26:177–210.
- Tripathi, B. D., and A. R. Upadhyay. 2003. Dairy effluent polishing by aquatic macrophytes. *Water, Air, and Soil Pollution* 143:377–385.
- Türker, O. C., C. Türe, H. Böcük, A. Çiçek, and A. Yakar. 2016. Role of plants and vegetation structure on boron (B) removal process in constructed wetlands. *Ecological Engineering* 88:143–152.
- Viechtbauer, W. 2010. Conducting meta-analyses in R with the metafor package. *Journal of Statistical Software* 36:1–48.
- Wang, H., C. C. Huang, Y. Ge, Y. W. Mo, S. X. Zhu, and J. Chang. 2014. Variation in plant biomass and nitrogen removal in two wetland blocks with species composition. *International Journal of Environmental Technology and Management* 17:404–417.
- Wang, Y., H. Yang, C. Ye, X. Chen, B. Xie, C. Huang, J. Zhang, and M. Xu. 2013. Effects of plant species on soil microbial processes and CH₄ emission from constructed wetlands. *Environmental Pollution* 174:273–278.
- Wardle, D. A. 2016. Do experiments exploring plant diversity–ecosystem functioning relationships inform how biodiversity loss impacts natural ecosystems? *Journal of Vegetation Science* 27:646–653.
- Winfree, R., J. W. Fox, N. M. Williams, J. R. Reilly, and D. P. Cariveau. 2015. Abundance of common species, not species richness, drives delivery of a real-world ecosystem service. *Ecology Letters* 18:626–635.

- Zedler, J. B., and S. Kercher. 2005. Wetland resources: status, trends, ecosystem services, and restorability. *Annual Review of Environment and Resources* 30:39–74.
- Zhang, C. B., W. L. Liu, W. J. Han, M. Guan, J. Wang, S. Y. Liu, Y. Ge, and J. Chang. 2016. Responses of dissimilatory nitrate reduction to ammonium and denitrification to plant presence, plant species and species richness in simulated vertical flow constructed wetlands. *Wetlands*. <https://doi.org/10.1007/s13157-016-0846-4>
- Zhang, C. B., W. L. Liu, J. Wang, T. Chen, Q. Q. Yuan, C. C. Huang, Y. Ge, S. X. Chang, and J. Chang. 2011. Plant functional group richness-affected microbial community structure and function in a full-scale constructed wetland. *Ecological Engineering* 37:1360–1368.
- Zhang, C. B., W. L. Liu, J. Wang, Y. Ge, B. H. Gu, and J. Chang. 2012a. Effects of plant diversity and hydraulic retention time on pollutant removals in vertical flow constructed wetland mesocosms. *Ecological Engineering* 49:244–248.
- Zhang, C. B., H. Y. Sun, Y. Ge, B. J. Gu, H. Wang, and J. Chang. 2012b. Plant species richness enhanced the methane emission in experimental microcosms. *Atmospheric Environment* 62:180–183.
- Zhang, Z., Z. Rengel, and K. Meney. 2007. Nutrient removal from simulated wastewater using *Canna indica* and *Schoenoplectus validus* in mono-and mixed-culture in wetland microcosms. *Water, Air, and Soil Pollution* 183:95–105.
- Zhang, C. B., J. Wang, W. L. Liu, S. X. Zhu, H. L. Ge, S. X. Chang, C. Jie, and Y. Ge. 2010. Effects of plant diversity on microbial biomass and community metabolic profiles in a full-scale constructed wetland. *Ecological Engineering* 36:62–68.
- Zhao, Z., J. Chang, W. Han, M. Wang, D. Ma, Y. Du, Z. Qu, S. X. Chang, and Y. Ge. 2016. Effects of plant diversity and sand particle size on methane emission and nitrogen removal in microcosms of constructed wetlands. *Ecological Engineering* 95:390–398.
- Zhao, Z., H. Shi, Y. Liu, H. Zhao, H. Su, M. Wang, and Y. Zhao. 2014. The influence of duckweed species diversity on biomass productivity and nutrient removal efficiency in swine wastewater. *Bioresource Technology* 167:383–389.
- Zurita, F., J. De Anda, and M. A. Belmont. 2009. Treatment of domestic wastewater and production of commercial flowers in vertical and horizontal subsurface-flow constructed wetlands. *Ecological Engineering* 35:861–869.

SUPPORTING INFORMATION

Additional supporting information may be found online at: <http://onlinelibrary.wiley.com/doi/10.1002/eap.2074/full>

DATA AVAILABILITY

The database developed for the meta-analysis is available on Figshare: <https://doi.org/10.6084/m9.figshare.4958687>